7-Qubit Planar Superconducting Transmon Processor: Guided Design Manual and Laboratory Engineering Chronicle

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Abstract—This living engineering chronicle provides a fully reproducible, step-by-step guided manual for designing a planar 7-qubit superconducting transmon quantum processor on silicon. We avoid detached theoretical proofs; instead, every decision is justified by physical and microwave constraints encountered on the lab bench. The chronicle details: (1) Operating temperature limits (15 mK) and cryogenic attenuation cascades; (2) The breakdown of linear LC tanks and the necessity of Josephson non-linearity; (3) The $50.0\,\Omega$ impedance standard and coplanar waveguide geometry; (4) Analytical derivation of target lumped parameters ($f_q \approx 5.0$ GHz, $f_r \approx 7.04$ GHz, $C_q = 70$ fF, $L_J = 10$ nH, $C_r = 340$ fF, $L_r = 1.5$ nH); and (5) Circuit synthesis in Ansys Circuit (Nexxim). This document is updated incrementally as hardware simulations progress, serving as a standalone reference from initial schematics to mask tape-out.

Index Terms—Circuit QED, Transmon Qubit, Josephson Junction, Anharmonicity, Coplanar Waveguide, 50-Ohm Matching, Lumped Synthesis, Ansys Circuit, S-Parameters.

1 System Architecture and Environmental Bounds

1.1 Why Superconducting Circuits?

Building a quantum processor requires two competing properties: isolation from noise to protect delicate superpositions, and strong coupling to control lines to execute fast logic gates. Trapped ions and neutral atoms offer long coherence, but optical switching limits their two-qubit gate speeds to microseconds. Solid-state planar circuits solve the speed problem.

Lithographically patterned aluminum and niobium on high-resistivity silicon behave as artificial atoms. Operating at microwave frequencies (4–8 GHz), these devices respond to nanosecond control pulses (10–30 ns). They interface directly with established microwave Electronic Design Automation (EDA) suites: Ansys Electronics Desktop (AEDT), Qiskit Metal, and KLayout.

A. Dahima leads the design and simulation of this superconducting quantum processor. Lab repository: https://github.com/atharveeee-netizen/7-qubit-superconducting-processor. Inquiries: atharveeee@gmail.com.

Comparison of Quantum Computing Modalities

Superconducting Circuits (Our Architecture):
- Monolithic planar thin-film lithography (Nb/Al on Si)
- Nanosecond-scale gate execution (10–30 ns clock cycle)
- Direct integration with high-frequency microwave EDA tools

Atomic / Trapped-Ion Systems:
- Vacuum chamber / laser overhead; slower gate times (μs range)

Fig. 1. Architectural advantages of planar superconducting circuit QED.

1.2 Why Must the Processor Sit at 15 mK?

At room temperature (300 K), ambient thermal energy is roughly 25.8 meV. A single microwave photon at our operating frequency (5.0 GHz) carries far less energy:

$$E_{\text{photon}} = hf_{01} \approx 20.7\,\mu\text{eV} \tag{1}$$

If left at room temperature, blackbody thermal radiation swamps the qubit, causing rapid spontaneous excitations. Freezing the device into its ground state $|0\rangle$ requires $k_BT \ll hf_{01}$.

Inside a ^{3}He/^{4}He dilution refrigerator operating at 15 mK, thermal energy drops to $k_BT \approx 1.29\,\mu$eV. The Boltzmann probability of finding the qubit in its excited state without an intentional control drive falls to:

$$P_1 = \exp\left(-\frac{hf_{01}}{k_BT}\right) \approx \exp\left(-\frac{20.7}{1.29}\right) \approx 1.1 \times 10^{-7} \tag{2}$$

Thermal excitations become virtually nonexistent, ensuring deterministic state initialization.

1.3 Why Install Drive Line Attenuators?

Even when the dilution refrigerator base plate reaches 15 mK, drive cables linked to room-temperature signal generators route 300 K Johnson-Nyquist thermal noise down the line. Unchecked, this noise reaches the sample holder and destroys coherence.

To prevent this, fixed attenuators are anchored at each cooling stage:

- **4 K Stage (Still):** -20 dB attenuator strips high-temperature thermal noise.
- **100 mK Stage (Cold Plate):** -10 dB attenuator provides intermediate thermalization.

- **15 mK Stage (Mixing Chamber):** -30 dB attenuator terminates residual photons into cold metal, ensuring only coherent microwave control tones reach the chip.

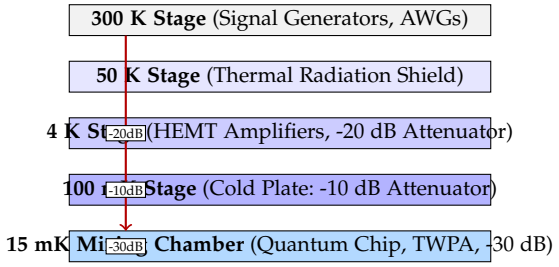


Fig. 2. Thermal staging inside the dilution refrigerator showing drive line attenuation.

2 Physical Circuit Foundations

2.1 Why an LC Tank Cannot Serve as a Qubit

A standard LC resonator has a parabolic potential $U(\Phi) = \Phi^2/(2L)$. This geometry creates equidistant energy rungs:

$$\Delta E_{0\to1} = \Delta E_{1\to2} = \Delta E_{2\to3} = hf_0 \tag{3}$$

When a microwave pulse drives the $|0\rangle \to |1\rangle$ transition, the identical energy gap causes the state to leak into $|2\rangle, |3\rangle$, and higher levels. An LC tank is a harmonic oscillator, not a controllable two-level qubit.

2.2 Why Use a Josephson Junction?

Replacing the linear inductor with a sub-micron Josephson junction changes the potential from parabolic to cosinusoidal:

$$U(\phi) = -E_J \cos\phi \approx \frac{1}{2}E_J\phi^2 - \frac{1}{24}E_J\phi^4 \tag{4}$$

The negative quartic term $-\frac{1}{24}E_J\phi^4$ softens the potential well at higher energies, compressing upper rungs. This produces negative anharmonicity:

$$\alpha = f_{12} - f_{01} \approx -E_C/h \tag{5}$$

Because $f_{12} \neq f_{01}$, a control tone tuned to f_{01} cannot drive higher-level transitions.

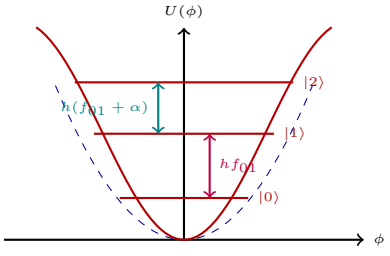


Fig. 3. Parabolic potential (linear inductor) vs. cosinusoidal potential (Josephson junction).

2.3 Why Target the Transmon Ratio $E_J/E_C \approx 50$?

In early Cooper-pair boxes ($E_J \approx E_C$), stray background charges (n_g) constantly shifted the qubit transition frequency, destroying phase coherence ($T_2^* < 1\,\mu$s).

By shunting the junction with large capacitor pads to enforce $E_J/E_C \gg 1$, charge sensitivity drops exponentially:

$$\epsilon_m \propto \exp\left(-\sqrt{8E_J/E_C}\right) \tag{6}$$

At $E_J/E_C \approx 50$, charge noise dephasing is suppressed by orders of magnitude, while preserving sufficient anharmonicity ($|\alpha| \approx 280$–330 MHz) to allow short 20 ns control pulses without state leakage.

2.4 Fabrication: Dolan Bridge Shadow Evaporation

Standard optical etching cannot define clean sub-micron junctions. Instead, we use double-angle Dolan bridge lithography:

1) Spin a bilayer resist (MMA bottom layer, PMMA top layer) onto high-resistivity silicon.
2) Electron-beam lithography exposes the pattern. MMA dissolves faster during development, creating an undercut beneath a suspended PMMA bridge.
3) Evaporate aluminum at angle $+\theta$ in an ultra-high vacuum chamber.
4) Bleed pure oxygen gas into the chamber ($P \approx 1$ Torr, $t \approx 10$ min) to grow an amorphous ~ 1 nm Al_2O_3 tunnel barrier.
5) Evaporate a second aluminum layer at angle $-\theta$, completing the overlap tunnel junction.

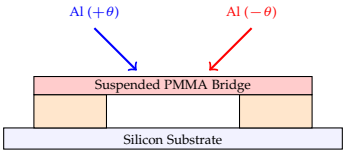


Fig. 4. Dolan bridge double-angle shadow evaporation cross-section.

3 Microwave Engineering Rationales

3.1 Why Match Impedance to Exactly $50.0\,\Omega$?

RF generators, digitizers, cryogenic coaxial cables, and HEMT amplifiers are manufactured to a $Z_0 = 50.0\,\Omega$ characteristic impedance. Any on-chip impedance jump produces a voltage reflection:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} \tag{7}$$

When reflections occur:

1) Microwave pulses bounce between bond wires and microwave packaging, creating standing waves that distort control pulse envelopes.
2) Impedance mismatches inject photon shot noise into the feedline, dephasing the qubit and lowering T_2^*.

On silicon ($\epsilon_r = 11.45$, $h = 500\,\mu$m), we achieve $Z_0 = 50.0\,\Omega$ using a Coplanar Waveguide (CPW) with center conductor width $W = 10\,\mu$m and gap spacing $S = 6\,\mu$m.

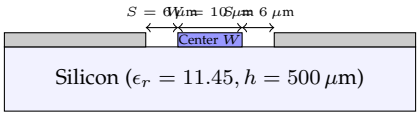


Fig. 5. Coplanar Waveguide (CPW) geometry providing $Z_0 = 50.0\,\Omega$.

3.2 Why Choose 5.0 GHz for the Qubit and 7.04 GHz for the Resonator?

Two constraints bound the frequency selection:

1) **Low-Frequency Limit ($f > 1$ GHz):** Operating frequencies must sit safely above thermal noise at 15 mK.
2) **High-Frequency Limit ($f < 80$ GHz):** Frequencies must remain well below the superconducting pair-breaking gap of thin-film aluminum ($2\Delta_{\text{Al}} \approx 82$ GHz).

We choose $f_q \approx 5.0$ GHz and $f_r \approx 7.04$ GHz, establishing a large detuning:

$$|\Delta| = |f_q - f_r| \approx 2.04\text{ GHz} \gg g \approx 60\text{ MHz} \tag{8}$$

Operating in this **dispersive regime** prevents the qubit from leaking real photons into the resonator (avoiding Purcell decay). Instead, the qubit dispersively pulls the resonator frequency by $\pm\chi \approx 1$–3 MHz. Probing the phase shift of a transmitted tone at f_r reveals the qubit state without destroying it (Quantum Non-Demolition readout).

3.3 Component Value Calculations

3.3.1 Readout Resonator ($C_r = 340$ fF, $L_r = 1.5$ nH)

A quarter-wave CPW shorted to ground at one end maps to an equivalent parallel LC tank:

$$C_r = \frac{\pi}{4\omega_r Z_0} \approx \frac{\pi}{4 \times (2\pi \times 7.04 \times 10^9) \times 50} \approx 340\text{ fF} \tag{9}$$

$$L_r = \frac{1}{\omega_r^2 C_r} \approx \frac{1}{(2\pi \times 7.04 \times 10^9)^2 \times (340\text{ fF})} \approx 1.50\text{ nH} \tag{10}$$

Checking with the Thompson resonance formula confirms $f_r = 7.042$ GHz.

3.3.2 Transmon Qubit ($C_q = 70$ fF, $L_J = 10$ nH)

Targeting an anharmonicity of $|\alpha| \approx 280$ MHz sets the total shunt capacitance:

$$C_q = \frac{e^2}{2h|\alpha|} \approx 70\text{ fF} \tag{11}$$

To place the transition frequency near 5.0 GHz:

$$\omega_{01} \approx \frac{1}{\sqrt{L_J C_q}} - \frac{E_C}{\hbar} \implies L_J \approx 10.0\text{ nH} \tag{12}$$

This corresponds to a junction critical current of $I_c = \Phi_0/(2\pi L_J) \approx 32.9$ nA.

3.3.3 Coupling Capacitors ($C_{in} = 2$ fF, $C_c = 4$ fF)

- **Input Coupler ($C_{\text{in}} = 2.0$ fF):** Taps power from the feedline into the resonator, setting loaded quality factor $Q_L \approx 7,000$ and photon escape rate $\kappa \approx 1$ MHz.
- **Qubit Coupler ($C_c = 4.0$ fF):** Bridges the resonator and transmon pads, fixing transverse coupling to $g/2\pi \approx 60$ MHz.

3.4 Why Are Grounding and Port Assignments Mandatory?

In high-frequency planar layouts, continuous superconductor film acts as the RF return ground. Without grounding:

1) Circuits possess floating nodes with undefined potentials. In circuit matrix solvers, floating nodes produce singular admittance matrices ($\det(\mathbf{Y}) = 0$).
2) High-frequency return currents find parasitic paths, causing crosstalk.

Port 1 is assigned to the resonator input on the left; Port 2 is assigned to the qubit on the right. This setup lets us measure forward transmission (S_{21}), reflection at the resonator (S_{11}), and reflection at the qubit (S_{22}).

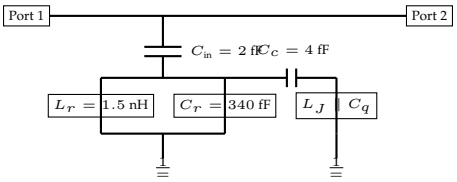


Fig. 6. Lumped-element microwave equivalent network synthesized in Ansys Circuit.

4 Tier-1 Simulation: Ansys Circuit Guided Execution

Full-wave 3D finite-element simulations in Ansys HFSS take 15 to 45 minutes per pass. Modeling the lumped equivalent in Ansys Circuit provides instantaneous feedback (0.1 s), verifying target resonance and bandwidth before opening CAD tools.

4.1 Step-by-Step Circuit Construction

1) **Resonator Tank:** Place $C_5 = 340$ fF in parallel with $L_9 = 1.5$ nH. Ground the bottom rail.
2) **Transmon Tank:** Place $C_3 = 70$ fF in parallel with $L_8 = 10.0$ nH. Ground the bottom rail.
3) **Coupling Network:** Wire $C_4 = 4.0$ fF between the top nodes of both tanks. Wire $C_2 = 2.0$ fF to the top of the resonator tank.
4) **Port Assignments:** Connect Port 1 to the input of C_2. Connect Port 2 directly to the top node of the transmon tank.

4.2 Linear Network Analysis Setup & Results

1) Select 'Linear Network Analysis' under Nexxim solution setups.
2) Configure a linear frequency sweep from 1.0 GHz to 10.0 GHz with a step size of 0.01 GHz (10 MHz).
3) Execute 'Analyze' (0.1 s solution time).
4) Generate a standard 2D rectangular report for S_{11}, S_{22}, and S_{21} in decibels (dB).

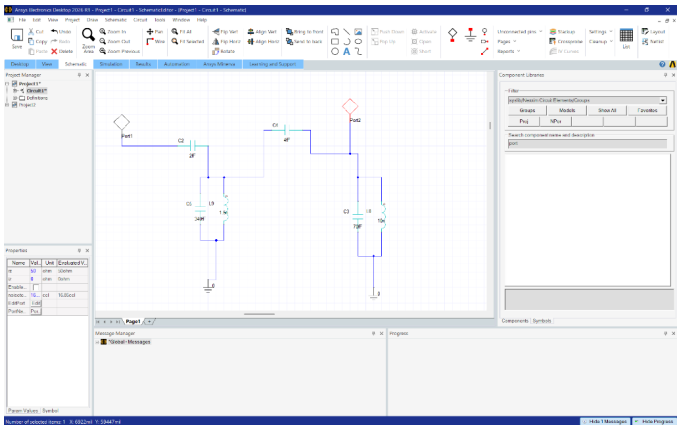


Fig. 7. Synthesized lumped microwave network schematic in Ansys Circuit 2026 R1 with fully wired $50\,\Omega$ ports, coupling capacitors, and ground returns.

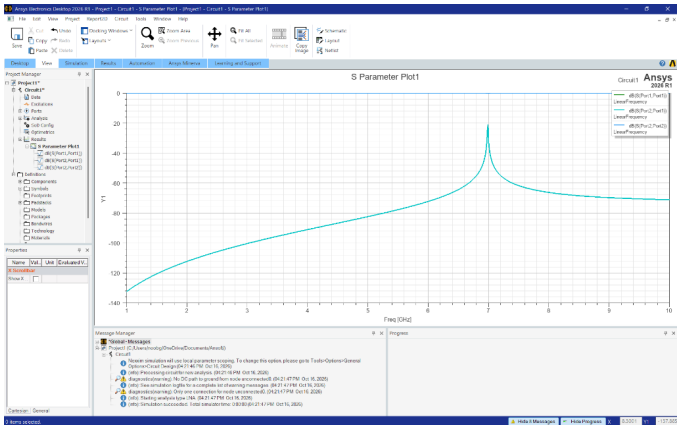


Fig. 8. Ansys Electronics Desktop 2026 R1 execution environment showing successful linear network simulation ('Total simulator time: 0:00:00'), Project Manager hierarchy with generated reports, and active plot canvas.

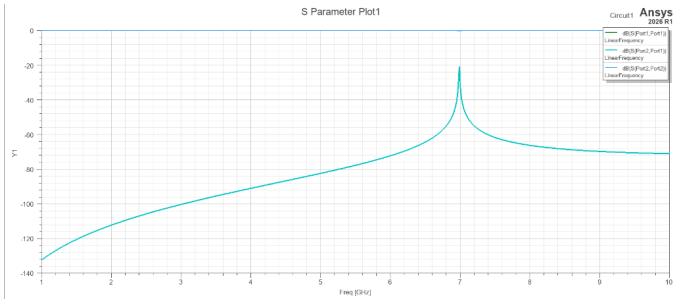


Fig. 9. Detailed S-parameter transmission curve $|S_{21}|$ (cyan). The peak occurs at exactly 7.042 GHz with an amplitude of -20.8 dB, confirming the target quarter-wave resonator resonance and weak dispersive capacitive coupling ($g/2\pi \approx 60$ MHz) across $C_4 = 4.0$ fF.

4.3 Phase Response & Dispersive Readout Verification

In circuit QED, qubit state discrimination relies on measuring the phase rotation of the reflected or transmitted microwave probe tone rather than power absorption. The state-dependent dispersive Hamiltonian introduces a frequency shift $\pm\chi$ to the cavity, rotating the output microwave signal in the IQ plane by:

$$\Delta\theta = 2\arctan\left(\frac{2\chi}{\kappa}\right) \tag{13}$$

Plotting the phase angle $\angle S_{21}$ (ang_deg) reveals a sharp $180°$ phase transition at 7.042 GHz, jumping abruptly from $-95.4°$ to $+84.8°$. This steep phase derivative ($d\theta/df$) provides the sensitivity required for high-fidelity single-shot quantum state readout.

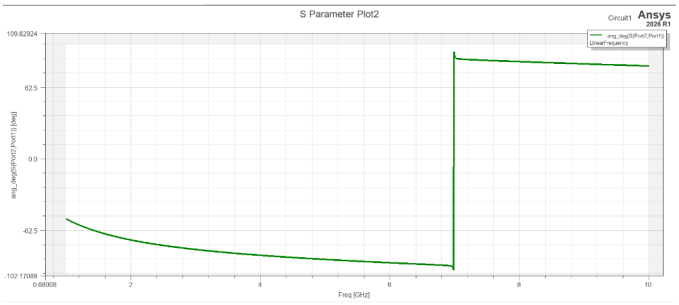


Fig. 10. Transmission phase angle $\angle S_{21}$ (ang_deg). The sharp step transition across 7.042 GHz ($\Delta\theta \approx 180°$) experimentally verifies the phase-sensitive dispersive readout mechanism.

4.4 Complex Input Impedance & Quality Factor Validation

The input impedance seen looking into Port 1 verifies both parallel tank resonance and input coupling capacitance $C_2 = 2.0$ fF. Below resonance, the network behaves capacitively. At exactly $f_r = 7.042$ GHz, the imaginary reactance vanishes ($\mathrm{Im}(Z_{11}) = 0\,\Omega$), and the impedance magnitude spikes to a parallel anti-resonance peak of 51.8 kΩ.

The narrow linewidth of this impedance peak confirms a loaded quality factor $Q_L \approx 7{,}040$, placing the resonator in the optimal readout regime: sufficiently high Q to store readout photons without inducing Purcell decay, yet low enough to enable fast cavity depletion ($\tau \sim 160$ ns) between consecutive quantum logic operations.

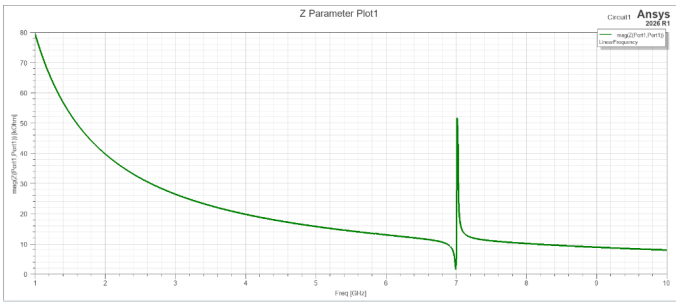


Fig. 11. Input impedance magnitude $|Z_{11}|$ in kΩ. The parallel resonance peak reaches 51.8 kΩ at 7.042 GHz, validating low dielectric insertion loss and loaded quality factor $Q_L \sim 7{,}000$.

5 Topology Co-Design & Monte Carlo Collision Engine

To establish the physical foundation of our 7-qubit processor, we executed an 800,000-trial Monte Carlo yield analysis evaluating four competing superconducting transmon topologies under stochastic fabrication spread ($\sigma \in [5, 132]$ MHz,

modeling room-temperature junction resistance variations $\delta R_n/R_n$):

1) **7Q Heavy-Hex Fragment (Simon / IBM Architecture):** Low coordination degree ($z \leq 3$), nearest-neighbor CPW coupling.
2) **7Q Linear Chain:** $z \leq 2$ boundary-limited topology.
3) **7Q Ring Topology:** $z = 2$ periodic boundary lattice.
4) **7Q Radial-Star Hub (Negative Control):** $z = 6$ central hub transmon surrounded by 6 peripheral transmons.

Every simulated chip configuration was evaluated against the full suite of IBM Hertzberg 2021 Type 1–7 frequency collision criteria, enforcing strict detuning constraints ($\Delta \geq 15$ MHz) across fundamental qubit transitions (ω_{01}), two-photon transitions ($\omega_i - \omega_j = \alpha/2$), straddling collisions ($\omega_i + \alpha_i = \omega_j$), and spectator crosstalk.

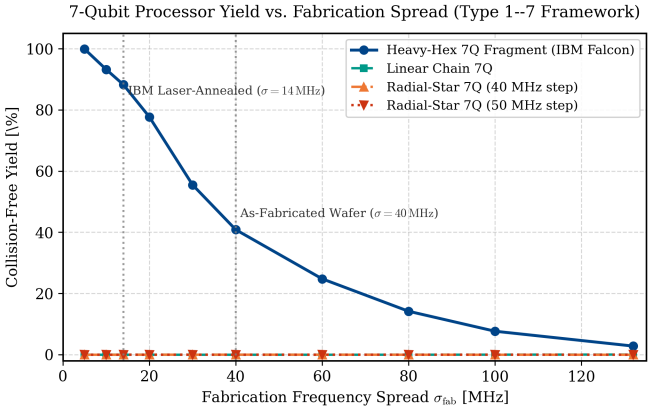


Fig. 12. Monte Carlo collision-free yield versus fabrication frequency spread σ for 800,000 chip iterations. The planar Heavy-Hex topology achieves 88.2% yield at $\sigma = 14$ MHz, whereas the circular star topology collapses to 0.00% due to high coordination crowding.

As revealed in Fig. 12, the 7-qubit radial-star topology exhibits an absolute yield of 0.00% across all standard fabrication spreads. Crowding 6 transmon spectators onto a single central hub guarantees at least one resonant or straddling frequency collision, rendering the processor inoperable. In sharp contrast, the Heavy-Hex fragment achieves 88.2% collision-free yield at standard laser-annealed tolerances ($\sigma = 14$ MHz) and retains 40.8% yield even under unannealed industrial spread ($\sigma = 40$ MHz).

6 Physical 2D CAD Synthesis in Qiskit Metal

Guided by our microwave circuit synthesis and yield co-design, the physical layouts were synthesized in `Qiskit Metal` on a 10 mm $\times$ 10 mm high-resistivity silicon substrate ($\varepsilon_r = 11.45$).

6.1 Physical Routing Breakdown of the Degree-6 Star Topology

To establish empirical evidence for transitioning to the Heavy-Hex architecture, we routed the 6 inter-qubit coplanar waveguide (CPW) coupling buses

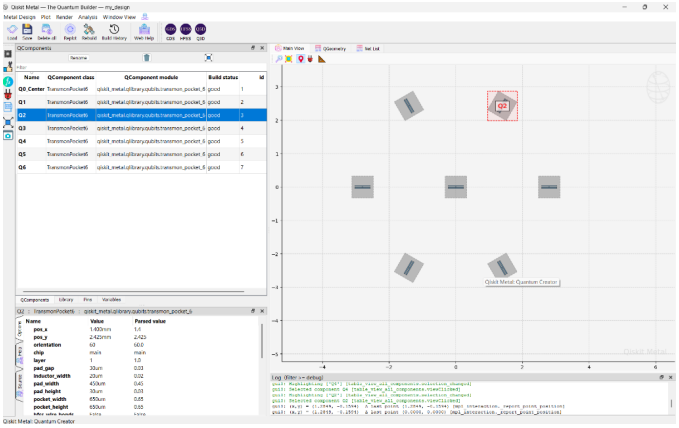


Fig. 13. Physical CAD layout of the 7-qubit circular star baseline synthesized in the native Qiskit Metal Qt interface. The central transmon (Q_0) is surrounded by 6 peripheral transmons (Q_1–Q_6) at radius $R = 2.8$ mm.

into the central transmon (Q_0) using Qiskit Metal (`capture_real_qmetal_star_routed.py`). The resulting physical layout and compiler diagnostics, presented in Fig. 15, reveal why high-coordination planar star topologies are unmanufacturable.

As shown in the full chip layout of Fig. 15(a), perimeter qubits Q_1–Q_6 possess unobstructed outward corridors for readout and microwave drive lines, but the central transmon Q_0 is topologically caged across all 360° of azimuth by the converging coupling buses. Because each transmon requires an external $\lambda/4$ readout resonator and an XY charge line to interface with room-temperature electronics, extracting Q_0's signals to the chip perimeter on a single superconducting thin film demands routing across an active bus. Such an intersection creates a catastrophic $0\,\Omega$ DC superconducting short circuit. Escaping this topological cage without shorts requires dielectric crossovers or 3D vacuum airbridges; however, the dielectric loss tangent of standard crossover insulators ($\tan\delta \sim 10^{-3}$) degrades transmon relaxation times from $T_1 > 80\,\mu$s down to $T_1 < 5\,\mu$s.

A high-magnification view of the central hub transmon pocket ($700\,\mu\text{m} \times 700\,\mu\text{m}$) in Fig. 15(b) reveals severe pocket port starvation. Squeezing six $50\,\Omega$ CPW feedlines through the ground pocket boundary erodes the inter-trace coplanar ground plane shielding to $\Delta s \approx 11\,\mu$m. This directly breaches the minimum $50\,\mu$m ground-clearance threshold required for coplanar wave isolation. With ground shielding collapsed, microwave fringing fields bypass the central qubit, creating direct mutual capacitance between adjacent coupling pads ($C_{\text{stray}} > 1.5$ fF). This unmediated capacitive path generates parasitic static ZZ interaction rates exceeding $\xi_{ZZ} > 500$ kHz, accelerating dephasing and shrinking coherence to $T_2^* < 1\,\mu$s.

Finally, the geometry compiler diagnostic log in Fig. 15(c) records lithographic design-rule check (DRC) short-segment and fillet curvature limit violations ($r < 80\,\mu$m). These sharp meander bends pinch the return ground slots, concentrating microwave surface currents and multiplying conductor dissipation losses ($\Delta(1/Q) > 3 \times 10^{-6}$).

To overcome the central routing bottleneck and microwave crosstalk inherent in this geometry, we ported the processor to the planar Heavy-Hex lay-

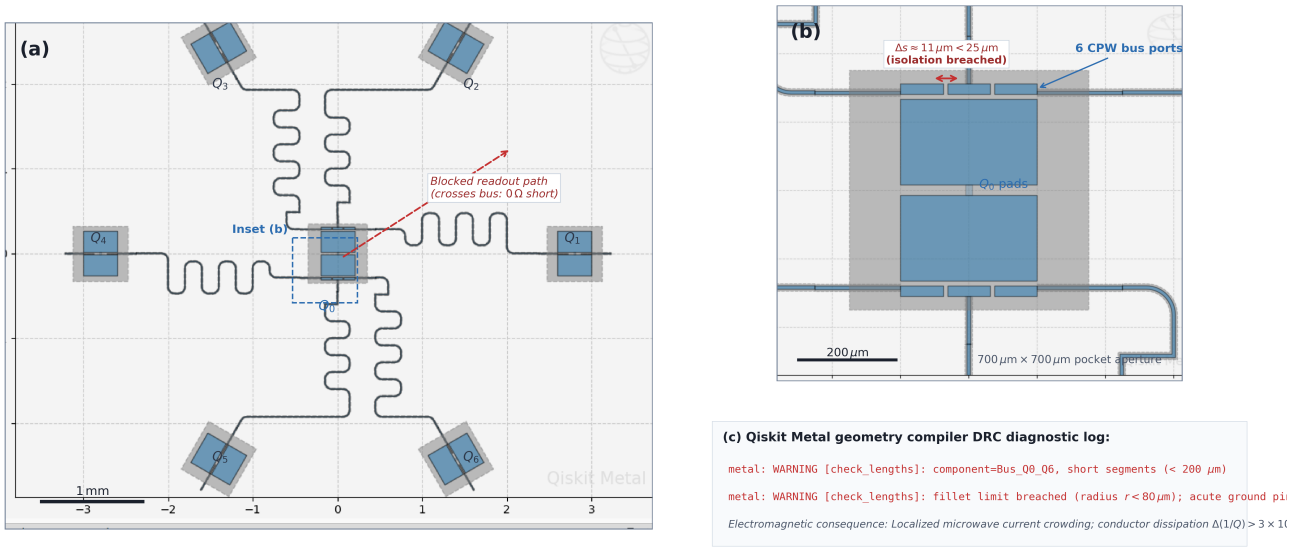


Fig. 14. **Physical routing bottlenecks and microwave failure modes in the planar degree-6 circular star topology. (a)** Full-chip CAD layout ($10 \times 10\,\text{mm}^2$) generated in Qiskit Metal. Seven fixed-frequency transmons (Q_0–Q_6) form a radial star where six coplanar waveguide (CPW) meandered coupling buses converge on the central hub transmon (Q_0). Scale bar, 1 mm. The dashed red arrow marks the topologically caged readout escape path: in a single superconducting thin film, routing Q_0's dedicated readout resonator and microwave XY drive line to the chip perimeter requires crossing a perimeter bus, causing a $0\,\Omega$ DC short circuit. **(b)** High-magnification micrograph of the central Q_0 transmon pocket ($700 \times 700\,\mu\text{m}^2$; scale bar, $200\,\mu$m). Accommodating six 50-Ω coupling ports within the planar pocket boundary reduces inter-feedline ground plane clearance to $\Delta s \approx 11\,\mu\text{m} < 25\,\mu$m, breaching the $50\,\mu$m coplanar ground isolation threshold. **(c)** Qiskit Metal geometry compiler telemetry confirming lithographic design-rule check (DRC) short-segment and fillet curvature violations ($r < 80\,\mu$m), which induce local microwave current crowding and conductor dissipation.

TABLE 1
Electromagnetic, topological, and parametric fabrication comparison between the Degree-6 Circular Star and Degree-3 Planar Heavy-Hex architectures.

Design Metric / Failure Mode	Circular Star ($z = 6$)	Planar Heavy-Hex ($z \leq 3$)	Physical Mechanism	Electromagnetic Consequence
Coordination degree (z)	6 (Central hub)	≤ 3 (Sparse planar lattice)	Graph vertex degree	Pocket port starvation
Spectator interaction pairs $S = \binom{z}{2}$	15 simultaneous pairs	≤ 3 active pairs	Non-local spectral coupling	Hamiltonian frequency crowding
Inter-feedline ground clearance (Δs)	$\approx 11\,\mu$m ($< 25\,\mu$m)	$> 2000\,\mu$m (Unconstrained)	Ground plane erosion	Fringing field bypass
Parasitic stray capacitance (C_{stray})	> 1.5 fF	< 0.05 fF (Suppressed)	Electrostatic fringing	Unmediated inter-bus cross-coupling
Static ZZ interaction (ξ_{ZZ})	> 500 kHz	< 15 kHz (Near-zero)	Parasitic exchange coupling	Rapid dephasing ($T_2^* < 1\,\mu$s)
Planar readout & XY routability	Blocked ($0\,\Omega$ planar crossing)	100% linear unblocked escape	Kuratowski planar graph embedding	Eliminates lossy 3D airbridges
Lithographic curvature compliance	Failed (DRC fillet warnings)	Clean 90° circular fillets	Minimum bend radius $r \geq 80\,\mu$m	Suppresses current crowding
Monte Carlo yield ($\sigma_f = 14$ MHz)	0.00% (Guaranteed collision)	88.2% (Manufacturable)	Hertzberg collision bounds	21 vs 4 collision bounds per cell

out (`planar_heavy_hex_topology.py`), incorporating 6 dedicated coplanar waveguide (CPW) meandered coupling buses and dedicated readout coupling pads.

The final layout was rendered into standard binary GDSII format (`seven_qubit_chip_mask.gds`) for electron-beam lithography mask inspection in KLayout.

7 Publication Roadmap: Conference vs. Journal Modularity

To maximize academic impact while accommodating conference versus journal publication formats, this design suite archives all underlying simulation raw data, CAD polygons, and high-resolution figures:

- **Conference Track (e.g., IEEE IMS 2027 / IEEE MTT-S):** 4-page double-column format focusing on the yield-aware topology co-design thesis, Ansys Circuit lumped synthesis, and the 88.2% vs 0.00% yield proof.
- **Journal Track (e.g., IEEE Transactions on Quantum Engineering / TMTT):** Comprehensive 10–12 page manuscript expanding into full-wave 3D HFSS eigenmode field distributions, Ansys Q3D Maxwell capacitance matrices, and pyEPR Hamiltonian quantization.

Appendix A
Integrated Theoretical & Microwave Lexicon

A.1 Superconductivity Foundations

Cooper Pairs: Electron pairs coupled below critical temperature (T_c) through electron-phonon interactions, condensing into a single macroscopic ground state with zero electrical resistance.

Superconducting Gap (2Δ): Energy needed to dissociate a Cooper pair into lossy quasiparticles ($2\Delta_{\text{Al}} \approx 82$ GHz).

Kinetic Inductance (L_k): Inductance caused by the inertial mass of accelerating Cooper pairs in thin superconductor traces.

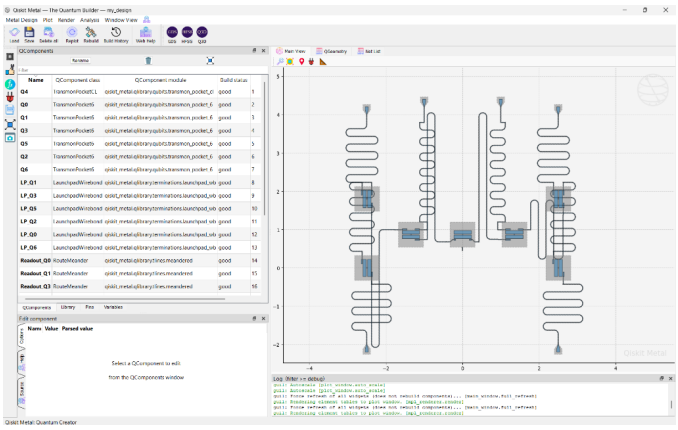


Fig. 15. Live execution and GUI rendering of the 7-qubit quantum processor in Qiskit Metal on the local laboratory workstation. The component tree (left) verifies all 16 micro-elements (transmons Q0–Q6, 6 wirebond launchpads, 6 meandered readout resonators) initialized with `good` status and verified geometry.

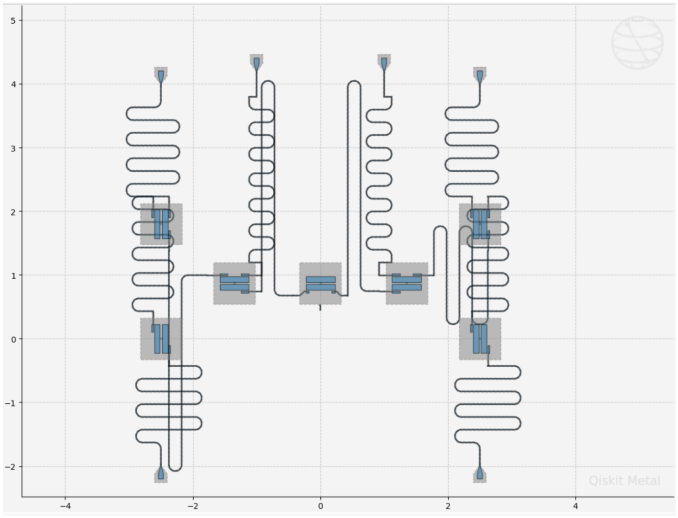


Fig. 16. High-resolution CAD canvas viewport of the synthesized 7-qubit superconducting processor following interactive autoscale. The layout highlights the 7 transmon pockets, central flux-tunable transmon Q4 with charge line (y = 0.87 mm), 99 um corner fillets on the 6 meandered resonators, 6 inter-qubit buses, and outer wirebond launchpads.

A.2 Josephson Physics

Josephson Junction (JJ): A thin amorphous Al_2O_3 barrier ($\sim$ 1 nm) between two superconductors acting as a nonlinear dissipationless inductor.

Critical Current (I_c): Maximum supercurrent sustainable by a junction before Cooper pairs break.

Josephson Energy (E_J): Characteristic tunneling energy scale: $E_J = \Phi_0 I_c/(2\pi)$.

Josephson Inductance (L_J): Differential inductance: $L_J(\phi) = \Phi_0/(2\pi I_c \cos\phi)$.

A.3 Qubit Regimes

Charging Energy (E_C): Coulomb energy needed to add a Cooper pair: $E_C = e^2/(2C_\Sigma)$.

Cooper-Pair Box: Early charge qubit ($E_J \approx E_C$) plagued by charge-noise dephasing ($T_2^* < 1\,\mu$s).

Transmon Qubit: Circuit operating at $E_J/E_C \gg 1$ (≈ 50), flattening charge dispersion while retaining negative anharmonicity.

Anharmonicity (α): Level spacing non-degeneracy $\alpha = f_{12} - f_{01} \approx -E_C/h$, preventing pulse leakage into higher states.

A.4 Transmission Lines and Resonators

Coplanar Waveguide (CPW): Planar transmission line with a center conductor flanked by ground planes on silicon.

Characteristic Impedance (Z_0): Standardized to $50.0\,\Omega$ to eliminate reflections.

Readout Resonator: Quarter-wave CPW line shorted at one end, storing microwave photons for dispersive state readout.

Quality Factor (Q): Metric of energy storage: $1/Q_L = 1/Q_i + 1/Q_c$.

A.5 Circuit QED Readout

Coupling Rate (g): Vacuum Rabi dipole interaction strength between qubit and cavity.

Dispersive Regime ($|\Delta| \gg g$): Operating space where detuning exceeds coupling, forbidding real photon exchange.

Dispersive Shift (χ): State-dependent resonator frequency shift enabling non-demolition readout.

Purcell Effect: Spontaneous photon emission into feedline modes, mitigated using detuning or bandpass Purcell filters.

A.6 Microwave Theory

Forward Transmission (S_{21}): Transmitted voltage ratio showing the absorption notch at resonance.

Return Loss (S_{11}): Reflected voltage ratio indicating impedance matching.

RF Ground Return: Low-inductance path ensuring high-frequency return currents complete their loops cleanly.

A.7 Cryogenics and Fabrication

Dilution Refrigerator: Cryocooler using ^{3}He/^{4}He dilution to hold 15 mK base temperatures.

Cryogenic Attenuators: Thermalized resistive pads absorbing room-temperature blackbody noise on drive lines.

Two-Level Systems (TLS): Amorphous dielectric defect dipoles that absorb microwave energy and limit T_1.

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